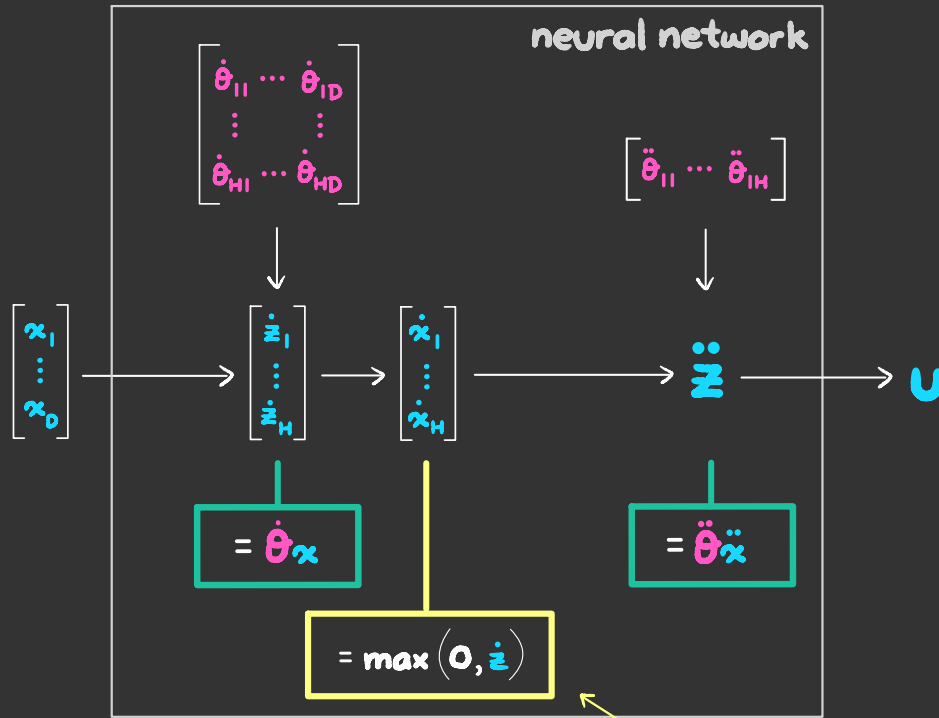
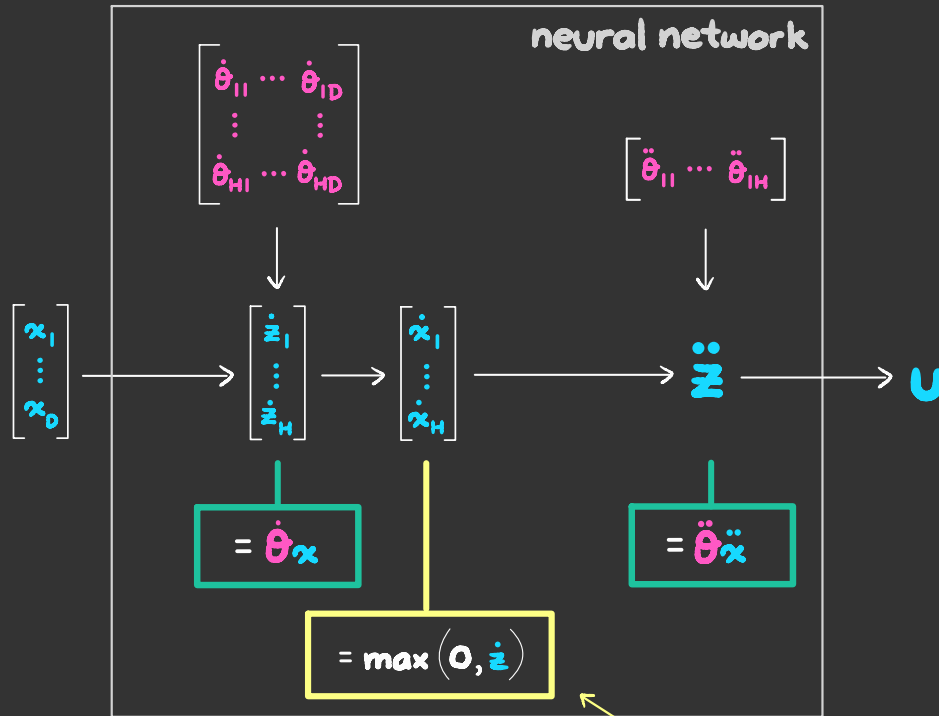


activation functions

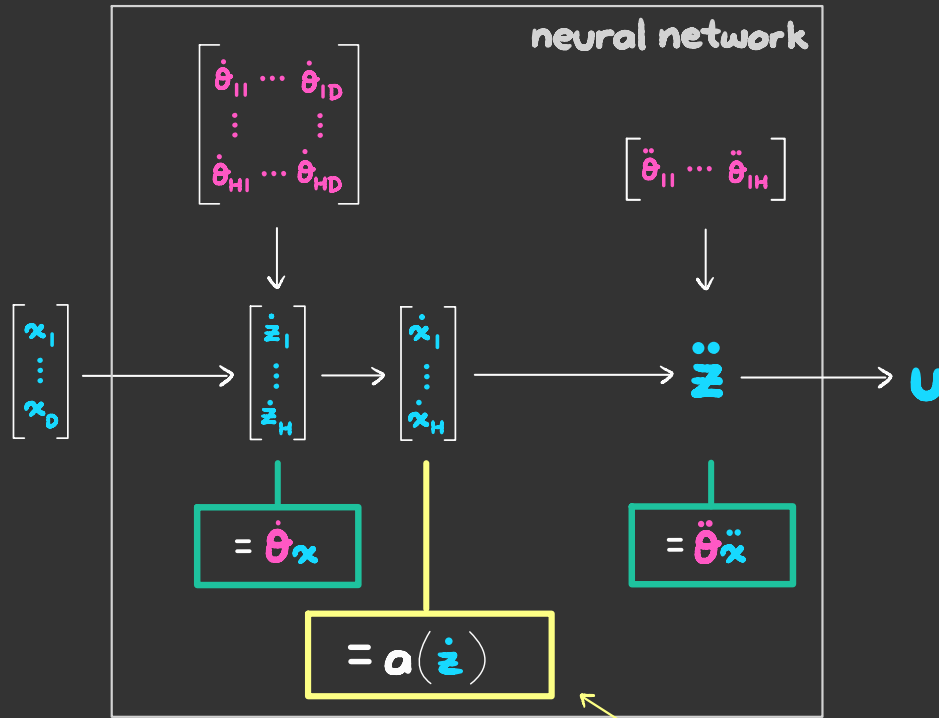
CSCI
270



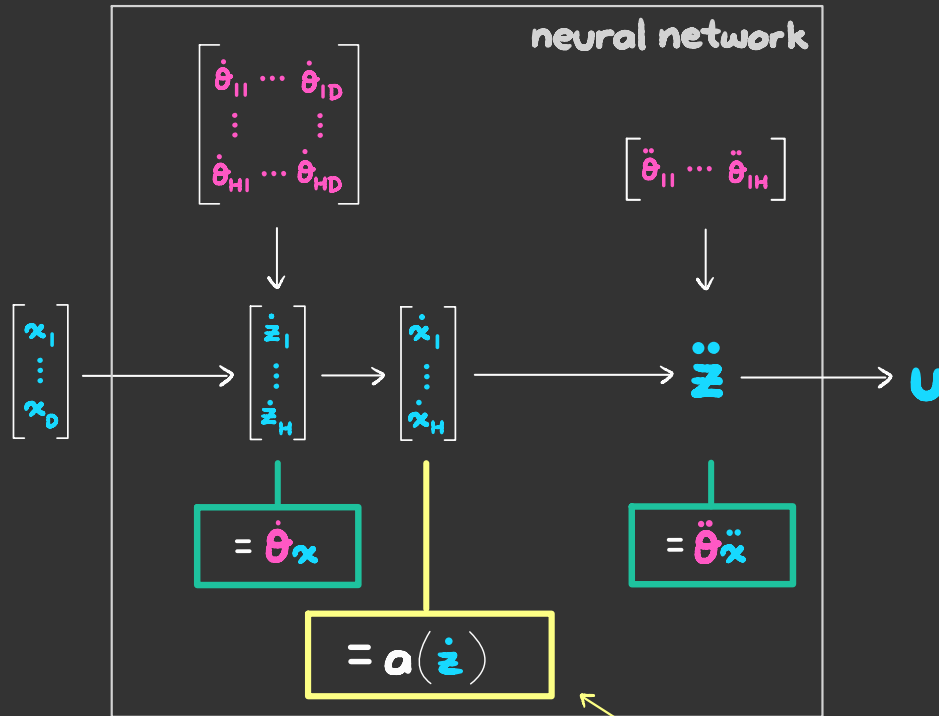
how important is **this step**?



let $a(\dot{z}) = \max(0, \dot{z})$

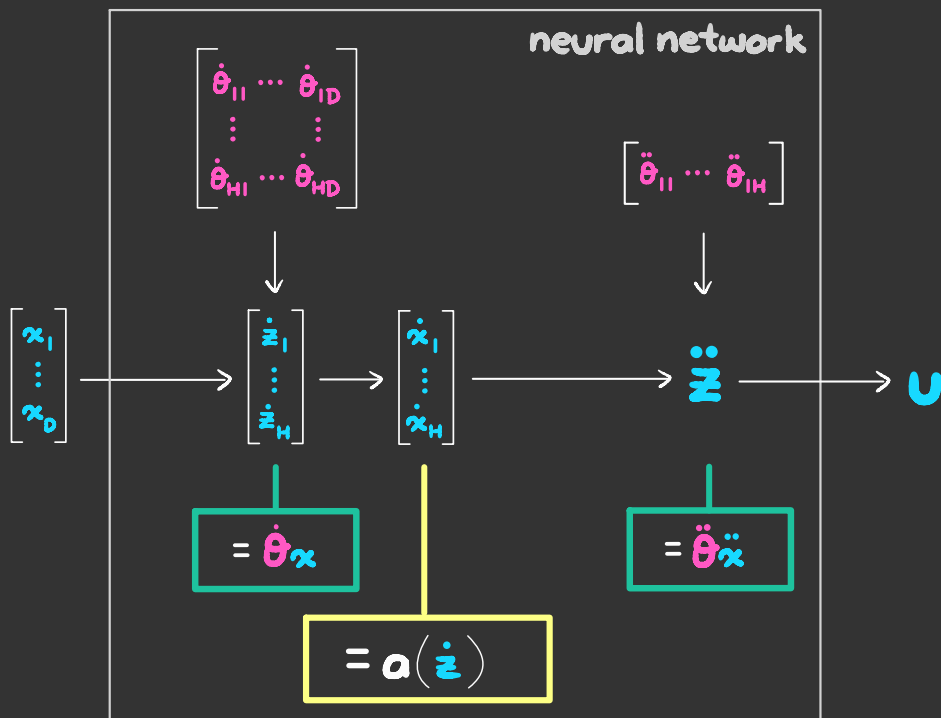


let $a(\dot{x}) = \max(0, \dot{x})$

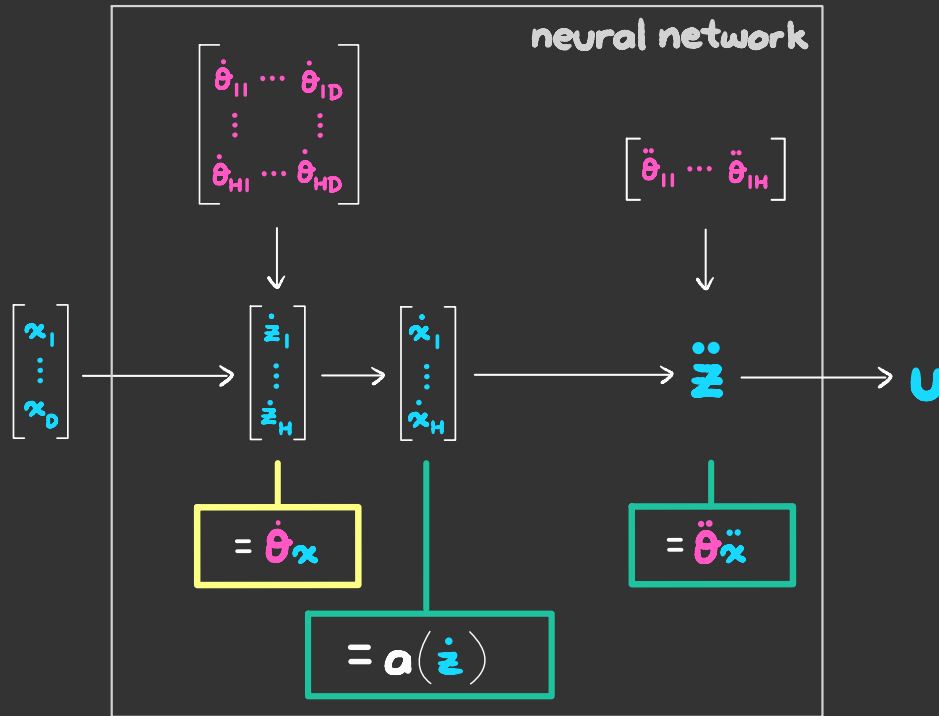


let $a(\dot{z}) = \max(0, \dot{z})$

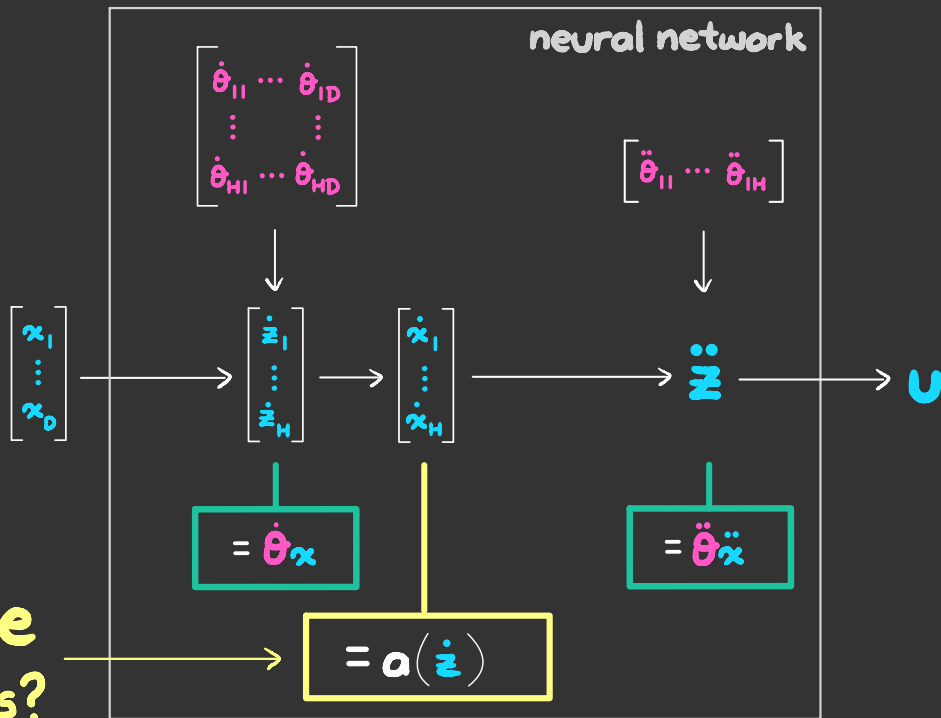




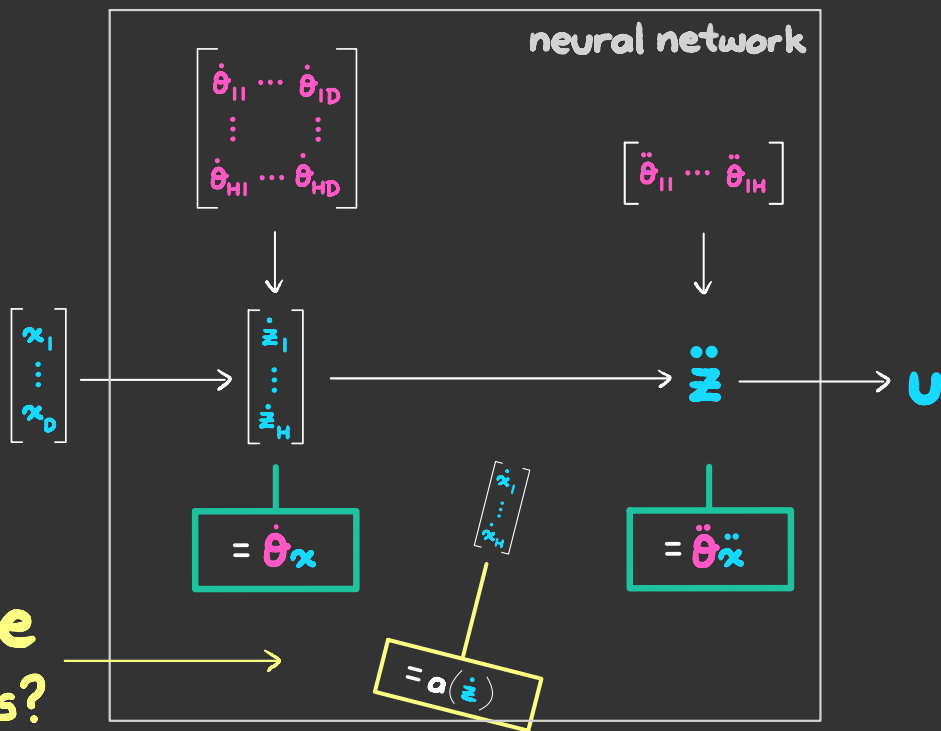
then: $\ddot{x} = \ddot{\theta} a(\dot{x})$



then: $\ddot{x} = \ddot{\theta} a(\dot{\theta} x)$



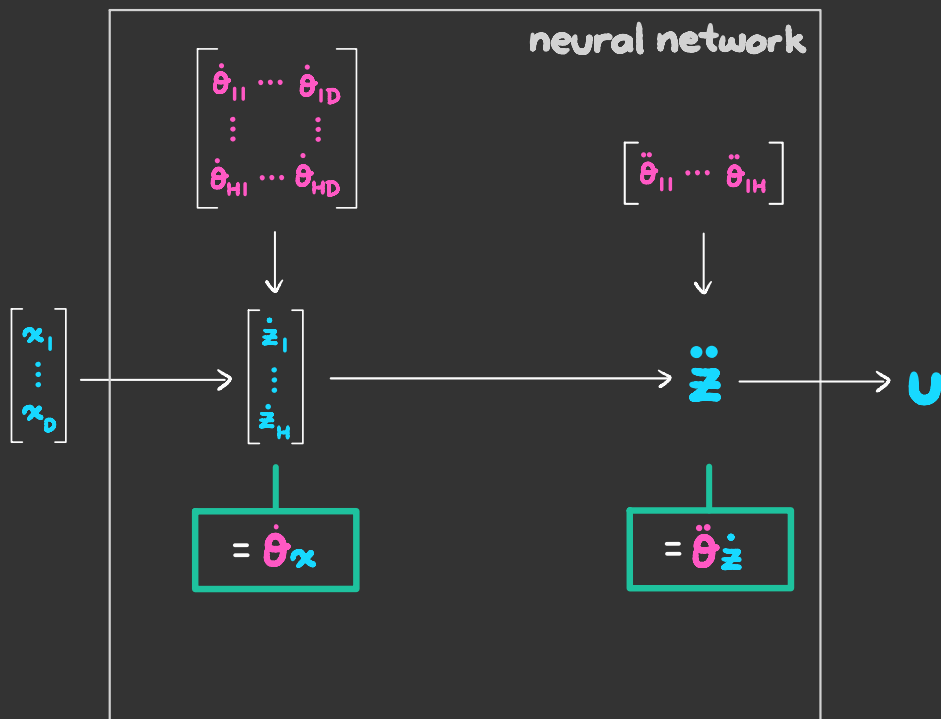
then: $\ddot{z} = \ddot{\theta} a(\dot{\theta} x)$



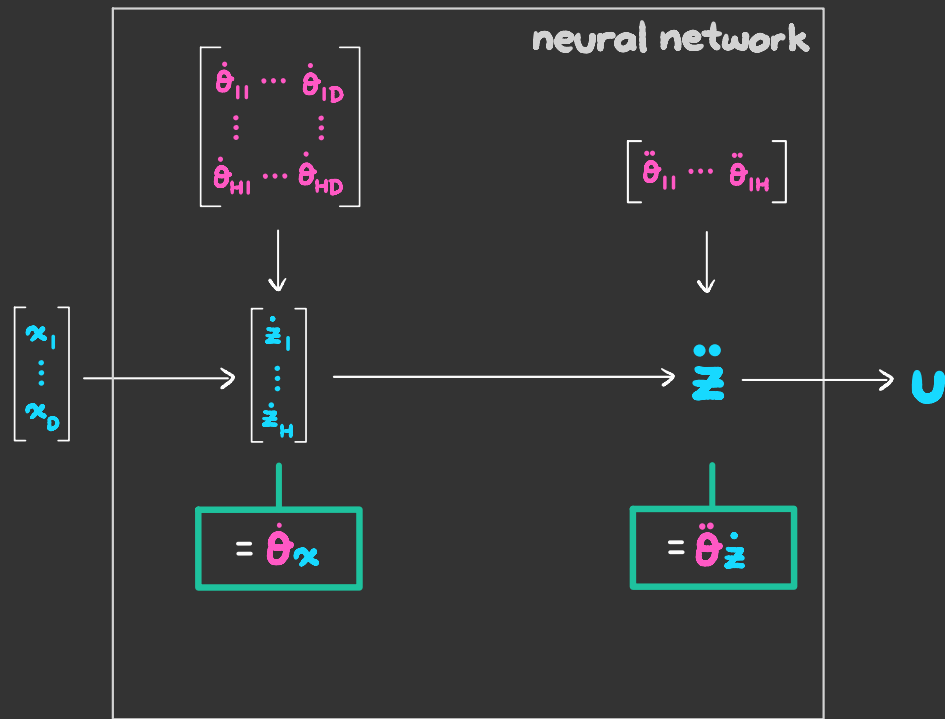
what if we
get rid of this?

then: $\ddot{z} = \ddot{\theta} \dot{x}$

$a(\cdot)$

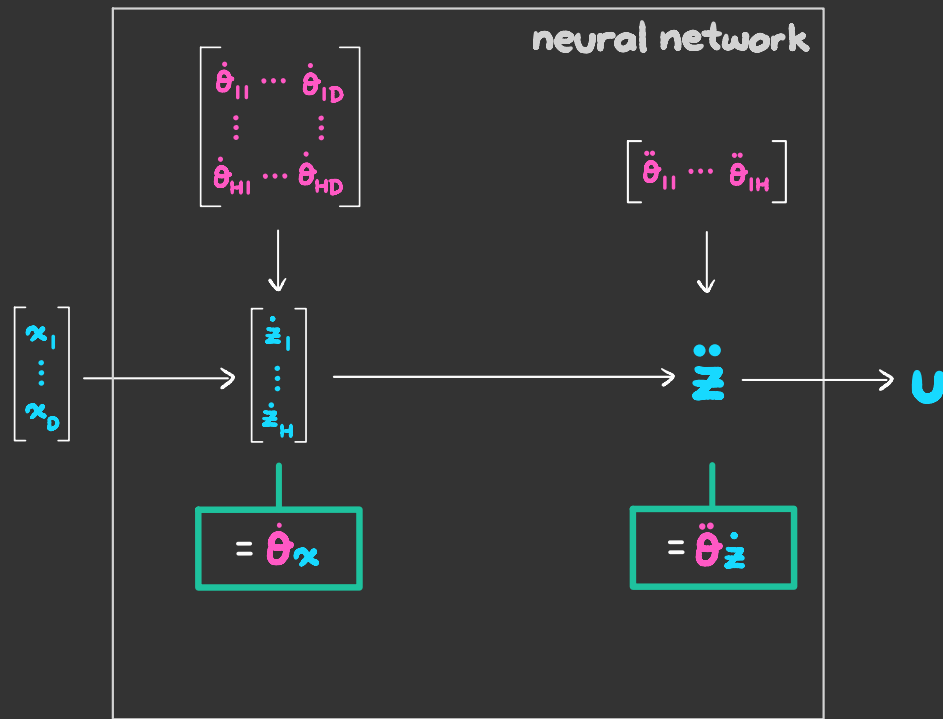


then: $\ddot{z} = \ddot{\theta} \dot{\theta} x$



but what
is this?

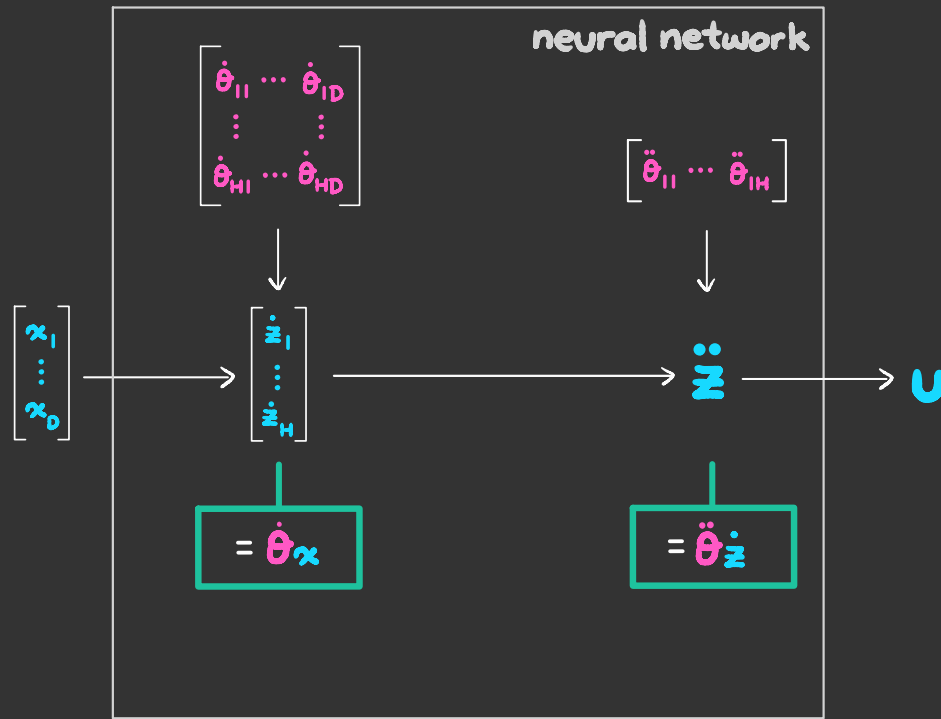
then: $\ddot{z} = \ddot{\theta} \dot{\theta} x$



but what
is this?

$$\begin{bmatrix} \ddot{\theta}_{11} & \dots & \ddot{\theta}_{1H} \\ \vdots & & \vdots \\ \ddot{\theta}_{H1} & \dots & \ddot{\theta}_{HH} \end{bmatrix} \times \begin{bmatrix} \dot{\theta}_{11} & \dots & \dot{\theta}_{1D} \\ \vdots & & \vdots \\ \dot{\theta}_{H1} & \dots & \dot{\theta}_{HD} \end{bmatrix} = \begin{bmatrix} \theta_{11} & \dots & \theta_{1D} \\ \vdots & & \vdots \\ \theta_{H1} & \dots & \theta_{HD} \end{bmatrix}$$

then: $\ddot{z} = \ddot{\theta} \dot{\theta} x$ ← just another matrix!



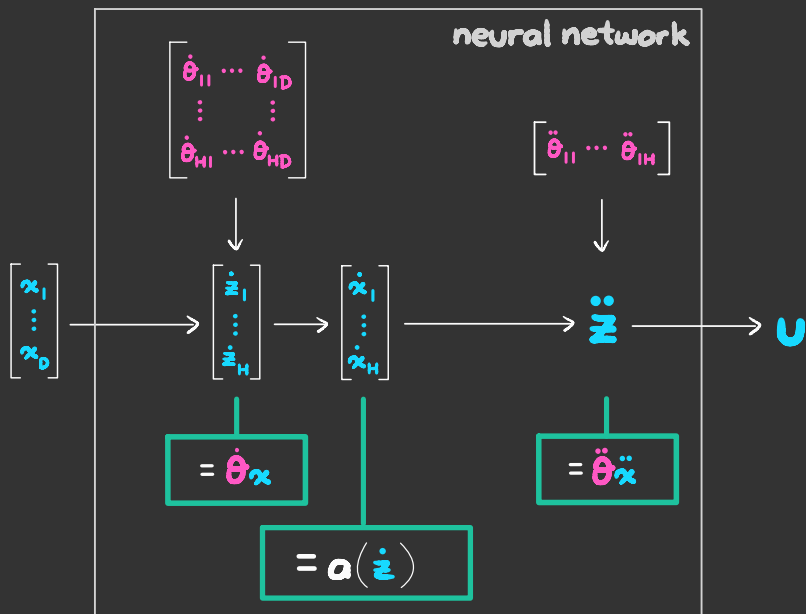
so the model
reduces to
linear/logistic
regression

then: $\ddot{z} = \ddot{\theta} x$

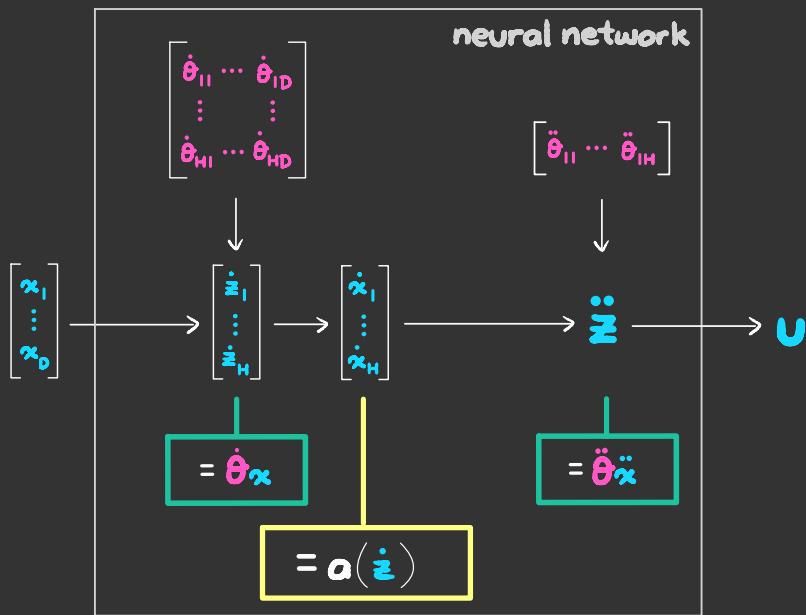


		instances					
		1	2	3	4	5	6
features	x_1 (bias)	1	1	1	1	1	1
	x_2 (height)	6.4	6.0	5.5	5.0	6.6	7.0
	x_3 (mass)	120	200	120	250	260	150
outcome	y (disease)	1	0	0	1	0	1

but linear/logistic
regression
could not explain
certain data

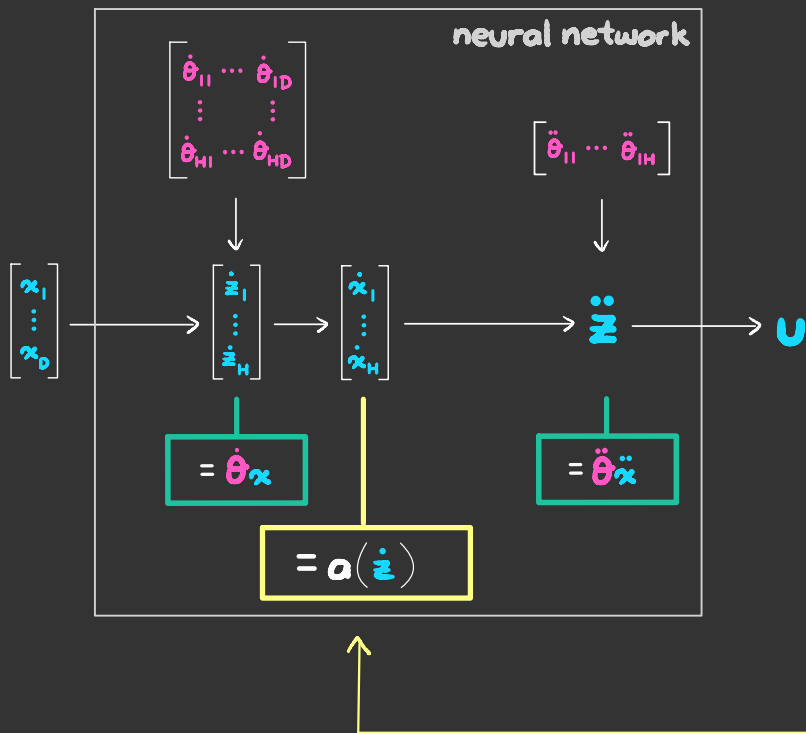


data that we
could explain with
a neural
network

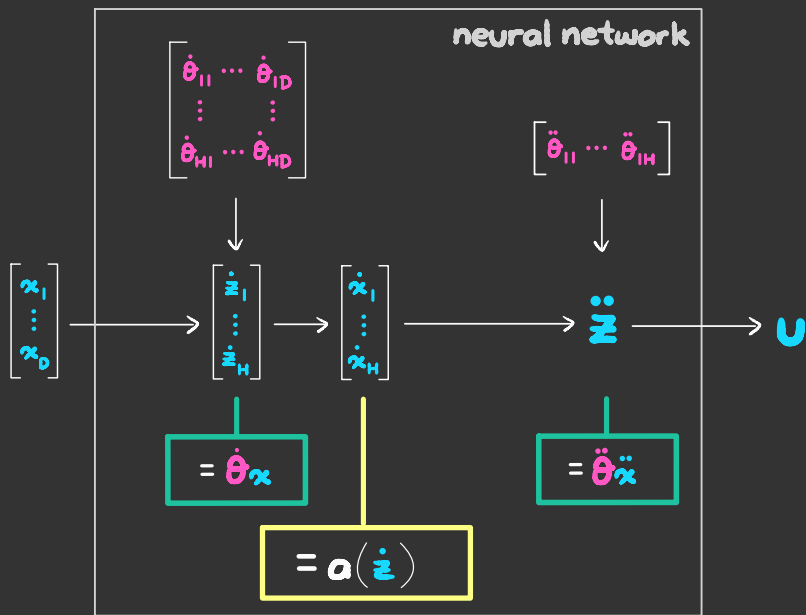


data that we
could explain with
a neural
network

so this must be important

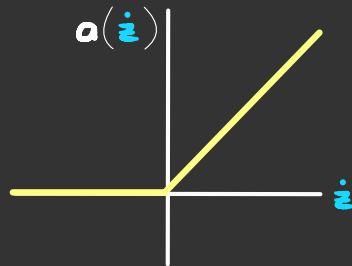


it is called an
activation
function



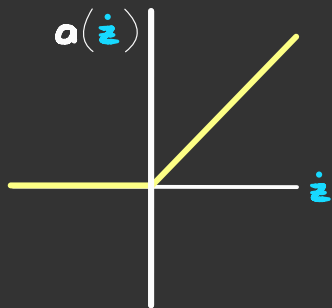
it has to be
nonlinear

e.g. $a(\dot{z}) = \max(0, \dot{z})$



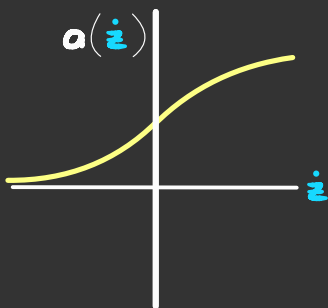
but we have options

relu



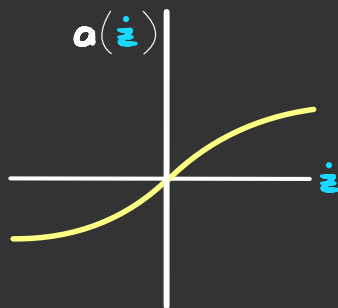
$$a(z) = \max(0, z)$$

sigmoid



$$a(z) = \frac{1}{1 + \exp(-z)}$$

tanh

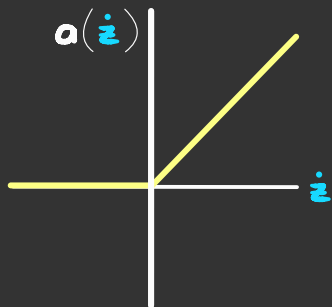


$$a(z) = 2\sigma(2z) - 1$$

currently popular, short for
rectified linear unit

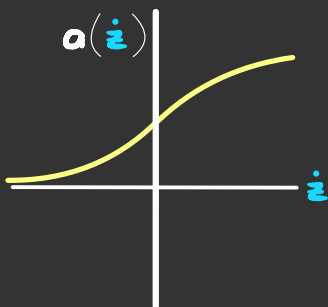
but we have options

relu



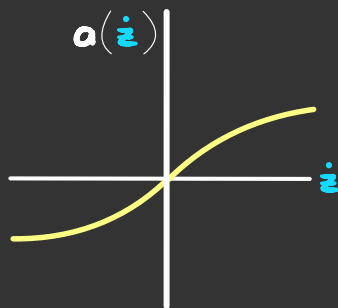
$$a(z) = \max(0, z)$$

sigmoid



$$a(z) = \frac{1}{1 + \exp(-z)}$$

tanh

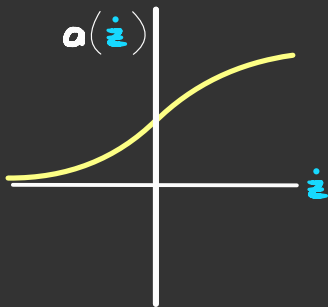


$$a(z) = 2\sigma(2z) - 1$$

historically popular

but we have options

sigmoid



$$a(z) = \frac{1}{1 + \exp(-z)}$$

why weren't neural nets popular prior to 2012?

1. our **labeled datasets** were thousands of times too small
2. our **computers** were millions of times too slow
3. we **initialized the weights** in a stupid way
4. we used the wrong type of **non-linearity**

Source: <http://www.androjkurenkov.com/writing/ai-a-brief-history-of-neural-nets-and-deep-learning-part-4/>

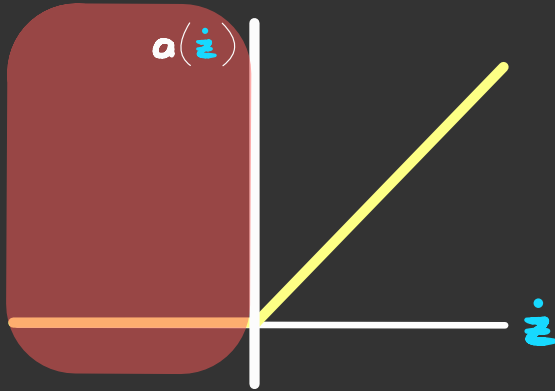


but not currently en vogue

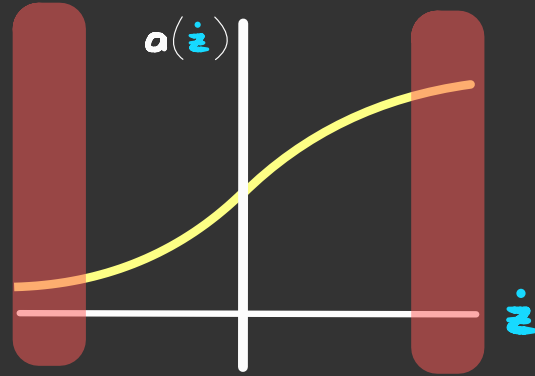
relu

vs

sigmoid



in the **red zone**,
big differences in z
have no impact on $a(z)$

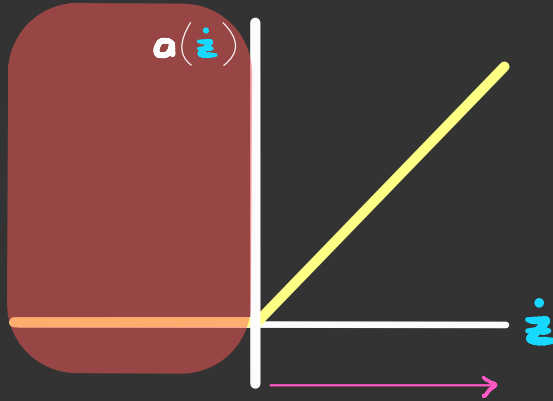


in the **red zones**,
big differences in z
have little impact on $a(z)$

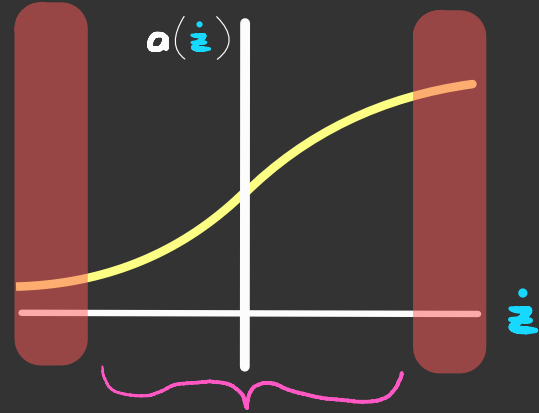
relu

vs

sigmoid



to have significant
impact on the result,
input needs to remain
positive



to have significant
impact on the result,
input needs to remain
in this **sweet** spot

a multilayer feedforward neural network
is just alternating **linear transforms**
and **nonlinear activation functions**

